

Pearson Edexcel International GCSE

(9-1) ICT: File Management Masterclass

File management involves the systematic organization, naming, storage, security, and version tracking of digital files. Proper file management saves time, prevents data loss, ensures you are working on the correct version of a product, and allows digital assets to be safely shared with others across various platform environments.

1. Directory Architecture & Folder Structures

A **folder structure** (or directory tree) is a hierarchical system used to organize files into logical, nested compartments on secondary storage media.

Hierarchical Folder Architecture

- **Root Directory:** The top-level folder on a storage device (e.g., Local Disk C:). All other folders and sub-folders branch out from this root.
- **Sub-folders:** Folders nested inside another folder. Grouping files into specialized sub-folders prevents clutter and streamlines retrieval.
- **Pathname:** The exact, sequential address pointing to a file's location within the directory tree.
 - *Example Path:* C:\School\ICT\Spreadsheets\PracticeActivity1.xlsx

2. Professional Naming Conventions & Version Control

Choosing vague, default filenames (such as Document1 or Logo-1) makes it difficult to find files later. Implementing a rigorous naming convention is essential for collaborative environments and long-term project management.

Sensible Filename Rules

A professional filename must be descriptive and consistent. It should ideally contain:

- The core **subject matter** or client name.
- The specific **asset description**.
- A standardized **version number** or date stamp.
 - *Poor Filename:* FinalLogo.png
 - *Sensible Filename:* School_Logo_v1.02.png

Advanced Version Management

To prevent accidentally overwriting previous iterations of a digital asset during development, users employ structured versioning codes:

- **MAJOR.REVISION.BUILDdate / BUILDnumber format:** A system where the filename reflects the scale of project modifications.
 - *Example:* E_Commerce_Project_v1.3_20260531
 - **Major changes** increase the first number (v1.0 to v2.0).
 - **Minor revisions** increase the second number (v1.2 to v1.3).
 - **Date stamps** allow users to quickly find the most recent file when sorting chronologically.

3. Data Protection: Saving Practices & Security

Digital files are vulnerable to hardware failures, power cuts, software crashes, and unauthorized access. Protecting files requires proactive habits and security configurations.

The Power of Regular Saving

- **Volatile Memory (RAM):** While you are actively working on a file, your modifications are held temporarily in RAM. If your computer suddenly loses power or crashes, all unsaved data in RAM is completely lost.
- **Secondary Storage:** Saving transfers data from volatile RAM to non-volatile secondary storage (like an internal SSD, external hard drive, or cloud storage).
- *Best Practice:* Use the universal keyboard shortcut **Ctrl + S** (Windows) or **Command + S** (Mac) frequently while working to commit changes to disk.

File Security Methods

To protect files from unauthorized modifications or exposure, apply these foundational security restrictions:

Security Method	Functional Mechanics	Practical Application
Password Protection	Encrypts the file contents, requiring a secret passcode string to decrypt and open it.	Protecting private company payroll spreadsheets or student grade lists.
Read-Only Status	Allows unauthorized users to safely open and view a file, but blocks them from making or saving any modifications to the original file.	Distributing a master school policy document or an employee handbook.
Access Rights / Permissions	Network rules that grant specific user groups different levels of access (Full Control, Modify, Read & Execute, or Deny).	Allowing managers to edit files while blocking general staff from seeing them.

4. Application vs. Published File Formats

File extensions (the three or four letters following the dot at the end of a filename, e.g., .docx, .pdf) declare a file's format. Formats are divided into two primary structural categories:

1. Application File Formats

These are the native, default working formats used by specific software packages while a digital product is actively being created or edited. They preserve specialized background data layers, formula paths, and development undo histories.

2. Published / Exported File Formats

When a digital product is completed, it is exported or "published" into a universal format. This ensures that anyone can view the finished file exactly as intended, even if they do not own the premium editing software. Published files are compressed and highly secure, but they lock content into a static layer, making it very difficult for others to modify.

Common Format Mappings

Application Type	Source Application	Native Application Format	Universal Published Format
Word Processor	Microsoft Word	.docx	.pdf (Portable Document Format)
Presentation	LibreOffice Impress	.odp	.ppsx (PowerPoint Show) / .pdf
Graphics Editor	GNU Image Manipulation (GIMP)	.xcf	.jpeg (Compressed) / .png (Supports transparency)
Web Authoring	Serif WebPlus	.wpp	.html (HyperText Markup Language)

Student Activities

Activity 1: Folder Tree Conception

1. Sketch a hierarchical folder layout designed to manage an entire academic year of schoolwork.
2. Ensure you establish logical sub-folders for separate subjects, and nested sub-folders within those subjects for different units (e.g., separate folders for Spreadsheets, Databases, and Word Processing).
3. Write down the complete system pathname for a file named ExamRevision.pdf saved within your ICT unit folder.

Activity 2: File Format Analysis

1. Create a basic text document in your native word processor.
2. Insert a shapes asset and a custom text heading, then save it normally as an editable application format (.docx or .odt).
3. Export or "Save As" the file into a Portable Document Format (.pdf).
4. Open the .pdf file using a web browser or basic viewer tool. Attempt to select a single word, move a shape asset, or delete a paragraph. Note down how published formats

protect content layouts from unauthorized editing compared to editable application formats.

Chapter Questions

1. Which one of these is the most sensible filename format for a logo asset designed for a school? A) Logo-1 B) FinalLogo C) School_Logo_v1.02 D) 102 School
2. Explain why it is vital for a computer user to save their active project files to secondary storage frequently. 3. State the reason why including versions in filenames is useful when developing a digital product. 4. State three distinct universal published or exported file formats. 5. Describe how an organization can use the MAJOR.REVISION.BUILDdate filename format to help employees quickly locate a specific asset iteration at a later date. 6. State one software method used to secure a sensitive spreadsheet from being edited by unauthorized personnel while still allowing them to view the data. 7. Define the term "Root Directory" within a hierarchical folder tree. 8. Explain why a graphics designer should export a company logo as a .png file format instead of a .jpeg file format if it needs to be placed over a dark website banner background. 9. Explain what happens to data held in RAM if a computer abruptly loses electrical power before the user hits the save command. 10. Which file extension represents a universal published layout format used to share text-heavy reports reliably across different operating system devices? A) .docx B) .xlsx C) .pdf D) .wpp

Answer Key & Explanations

1. **Correct Answer: C (School_Logo_v1.02)** *Explanation:* This option is descriptive, states the exact asset purpose, and implements strict version control numbers, making it easy to sort and identify.
2. **Core Reasons:** * Active modifications are held temporarily within volatile RAM, which requires continuous electrical power to maintain data.
- Saving transfers the file changes onto a stable, non-volatile secondary storage medium (like an internal hard drive), protecting your progress in case of a power outage, hardware failure, or application crash.
3. **Core Explanation:** * It creates a clear developmental timeline. It ensures team members always collaborate on the most up-to-date version, while preserving historical backup copies if a major error occurs and you need to roll back to a previous iteration.
4. **Valid Published Formats (Any three):** * .pdf (Portable Document Format)
- .jpeg / .jpg (Joint Photographic Experts Group image)
 - .png (Portable Network Graphics)
 - .html (HyperText Markup Language)
5. **Core Tracking Mechanics:** * The **Major** number groups files by milestone design phases.
- The **Revision** number exposes incremental tweaks within those major phases.
 - The **Build Date** allows users to sort files chronologically in file explorer windows, isolating the exact file built on a specific date without needing to open every document manually.
6. **Correct Answer: Read-Only Status** *Explanation:* Configuring a file to Read-Only status locks its editing capabilities. Users can safely open and read the data rows, but the software blocks them from overwriting or altering the master spreadsheet file.
7. **Correct Answer: Root Directory** *Explanation:* The root directory is the foundational master

folder at the very top of a storage device's file system, from which all other directories and sub-folders branch out.

8. Core Image Properties: * The .png file format supports **alpha-channel transparency layers**, allowing the background of the logo to remain invisible so it blends cleanly with the website banner.

- .jpeg does not support transparency and will force an opaque solid white box canvas boundary around the logo, ruining the website's design layout.

9. Core RAM Behavior: * RAM is highly volatile memory. When power is cut, its circuits lose their electrical state instantly, resulting in the permanent destruction of all unsaved data and active project modifications.

10. Correct Answer: C (.pdf) *Explanation:* The .pdf (Portable Document Format) is a universally compatible published format designed specifically to display structured text, layout formatting, and images identically across any platform or device.